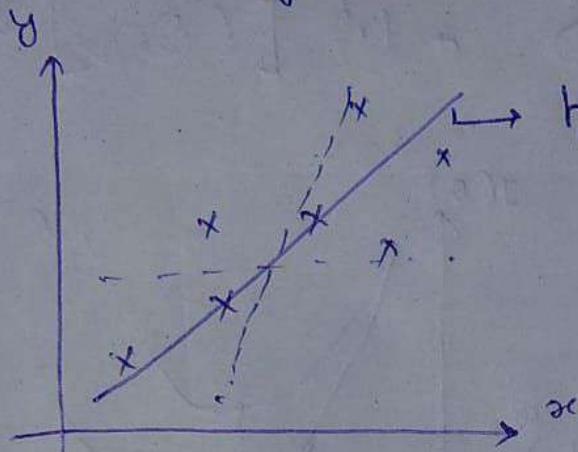


Ridge Regression, lasso

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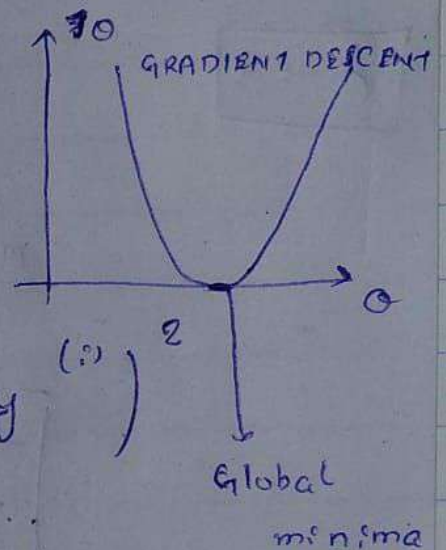
Regression, Elastic Regression

Linear Regression



$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

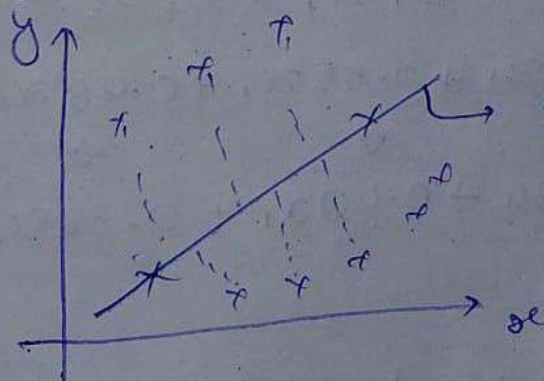
Independent feature



$$\text{cost fn} = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

Mean Squared Error

1) Ridge Regression (i.e. Regularization) → Reduce Overfitting



best fit line

Overfitting

Train data → Acc ↑ → Low Bias

Test data → Acc ↓ → High Variance

$A=0$ $d=1$
 $d=10$

$$\text{Cost fn} :- \frac{1}{2m} \sum_{i=1}^m (h_0(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{i=1}^n (\text{slope})^2$$

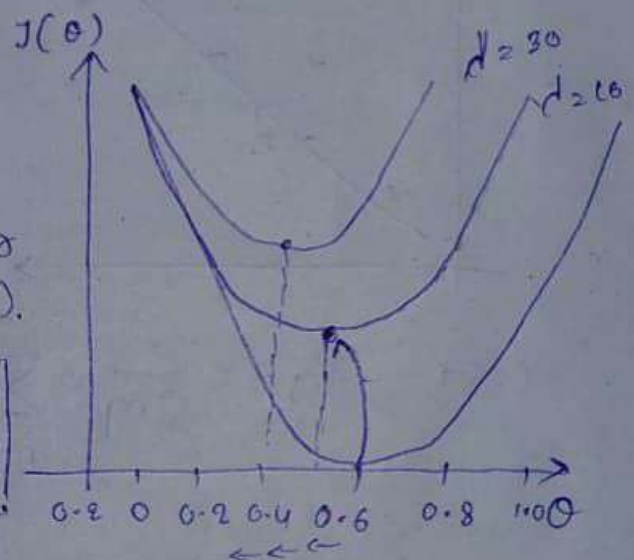
$$\lambda = 1$$

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Note: It will never be zero (0).

* Relation

$\lambda \uparrow$ slope $\downarrow$



λ = hyperparameter

$$\begin{aligned} x &\rightarrow 1 \\ y &\rightarrow 0.52 \end{aligned}$$

$$h_0(x) = 0.34 + 0.52x_1 + 0.48x_2 + 0.24x_3$$

$$= 0.34 + \underbrace{0.52}_{\downarrow} x_1 + \underbrace{0.48}_{\downarrow} x_2 + \underbrace{0.24}_{\downarrow} x_3$$

$$= 0.34 + 0.40x_1 + 0.38x_2 + \boxed{0.14x_3}$$

↑↑↑

Note: It reduce the impact or by reducing the coefficient that are not directly correlated by the output feature.